

Supplementary Table S1. List of Antibodies

Target	Conjugate	Species	Clone	Company
Human TruStain FcX				Biolegend
CD44	FITC	mouse	BJ18	Biolegend
TCR Vδ2	PE	mouse	B6	Biolegend
TCR Vδ2	Brilliant Violet 510	mouse	B6	Biolegend
TCR Vδ2	Biotin	mouse	B6	Biolegend
CD197 (CCR7)	PE/Dazzle™ 594	mouse	G043H7	Biolegend
CD27	PE/Cyanine7	mouse	M-T271	Biolegend
CD3	PerCP	mouse	SK7	Biolegend
CD62L	Alexa Fluor 647	mouse	DREG-56	Biolegend
CD127(IL-7Rα)	Alexa Fluor 700	mouse	A019D5	Biolegend
CD336 (NKp44)	APC/Cyanine7	mouse	P44-8	Biolegend
CD95 (Fas)	Brilliant Violet 605	mouse	DX2	Biolegend
CD279 (PD-1)	Brilliant Violet 650	mouse	NAT105	Biolegend
CD56 (NCAM)	Brilliant Violet 785	mouse	5.1H11	Biolegend
IL-2	FITC	mouse	MQ1-17H12	Biolegend
IL-4	PE/Dazzle™ 594	mouse	MP4-25D2	Biolegend
IL-17A	PE	mouse	BL168	Biolegend
TNF-α	APC	mouse	MAB11	Biolegend
IFN-γ	Brilliant Violet 421	mouse	4S.B3	Biolegend

Supplementary Table S2. List of primers

Gene	Forward	Reverse
Human-MIR155HG	CCACCCAATGGAGATGGCTC	AGTTGGAGGCAAAAACCCCT
Human-IL7R	ATCGCAGCACTCACTGACCTGT	TCAGGCACTTTACCTCCACGAG
Human-KLF3	CTCATGGTCTCCTTATCGGAGG	TGTCCTCTGTGGTTCGATCCCA
Human-WNT1	CTCTTCGGCAAGATCGTCAACC	CGATGGAACCTTCTGAGCAGGA
Human-IRF8	AGGTCTTCGACACCAGCCAGTT	GCACGAGAATGAGTTTGGAGCG
Human-FOXP1	CAAAGAACGCCTGCAAGCCATG	GGAGTATGAGGTAAGCTCTGTGG
Human-TNFSF8	CTCTGTGCCTTGTCTTCACGGT	CCTTTTCAGGATACATAAGAGGTC
Human-T-bet	ATTGCCGTGACTGCCTACCAGA	GGAATTGACAGTTGGGTCCAGG
Human-EMOES	AAATGGGTGACCTGTGGCAAAGC	CTCCTGTCTCATCCAGTGGGAA
Human-ROR gamma	GAGGAAGTGACTGGCTACCAGA	GCACAATCTGGTCATTCTGGCAG
Human-TCF7	CTGACCTCTCTGGCTTCTACTC	CAGAACCTAGCATCAAGGATGGG
Human-LEF1	AGCACAAACCTCTCAGGAGC	TCGAGTAGGAGGGTCCCTTG
Human-MYC	CCTGGTGCTCCATGAGGAGAC	CAGACTCTGACCTTTTGCCAGG
Human-MYB	GGGAACAGATGGGCAGAAATCG	GCTGGCTTTTGAAGACTCCTGC
Human-SELL	TCACAGTGTGCCTTCAGCTGCT	TCTGGTGCTGATAGAGGCTCAC
Human-TOX	CGCTACCTTTGGCGAAGTCTCT	CTGGCTCTGTATGCTGCGAGTT
Human-mTOR	AGCATCGGATGCTTAGGAGTGG	CAGCCAGTCATCTTTGGAGACC
Human-KLF2	CCAAGAGTTCGCATCTGAAGGC	CCGTGTGCTTTCGGTAGTGGC
Human-KLF4	CATCTCAAGGCACACCTGCGAA	TCGGTCGCATTTTTGGCACTGG
Human-GATA3	ACCACAACCACACTCTGGAGGA	TCGGTTTCTGGTCTGGATGCCT
Human-MIR155	TGCTAATCGTGATAGGGG	

Supplementary Table S3. List of Reagents and Chemicals

Reagents	Company	number
Zoledronic Acid (hydrate)	Caymanchem	CAS No#165800-06-6
FDA-Approved Drugs Screening Library	Caymanchem	Item No. #23538
Recombinant Human IL-2	PeptoThech	Cat. No#200-02
Dasatinib	MedChemExpress	Cat. No# HY-10181
microRNA RT Master Mix for qPCR III (poly A)	MedChemExpress	Cat. No# HY-K0512
Phorbol 12-myristate 13-acetate(PMA)	Sigma-aldrich	CAS No#16561-29-8
Ionomycin calcium salt	Sigma-aldrich	CAS No#56092-82-1
Brefeldin A	Sigma-aldrich	CAS No. #20350-15-6
BioTracker 655 Red Cytoplasmic Membrane Dye	Sigma-aldrich	Cat No. #SCT087
CFSE Cell Division Tracker Kit	Biolegend	Cat. No#423801
FITC Annexin V Apoptosis Detection Kit with 7-AAD	Biolegend	Cat. No#640922
MojoSort™ Streptavidin Nanobeads	Biolegend	Ca. Not#480016
Cyto-Last™ Buffer	Biolegend	Cat. No#422501
Cyto-Fast™ Fix/Perm Buffer Set	Biolegend	Cat. No#426803
Cell Staining Buffer	Biolegend	Cat. No#420201
Caspase-3/7 Detection Reagent	Thermofisher	Cat. No# C10434

Supplementary Table S4. Healthy donor (HD) used in this study

	Age	sex(M/F)	VD2 expansion (VD2/CD3%)
HD-1	41	M	80-90%
HD-2	36	F	80-90%
HD-3	35	M	30-60%
HD-4	20	F	80-90%
HD-5	42	M	40-70%
HD-6	20	F	65-90%
HD-7	20	M	50-85%
HD-8	52	F	80%
HD-9	44	M	25%
HD-10	40	M	70%
HD-11	20	F	80-90%
HD-12	25	F	70%-80%
HD-13	35	F	80-90%
HD-14	38	F	70%
HD-15	25	F	50%